

Exp.No -3: WireShark

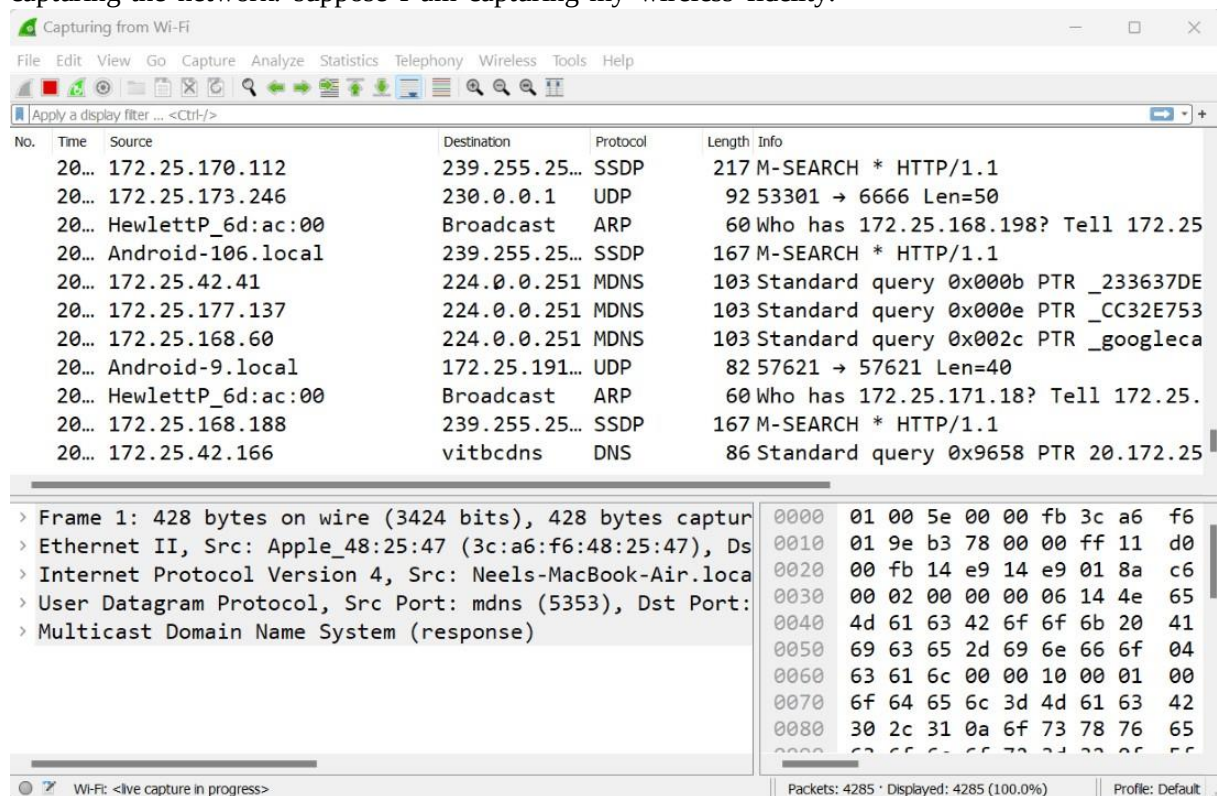
Password Capturing

Wireshark can capture not only passwords but any type of information transmitted over the network: usernames, email addresses, personal information, etc. As long as we can capture network traffic, Wireshark can sniff passing passwords.

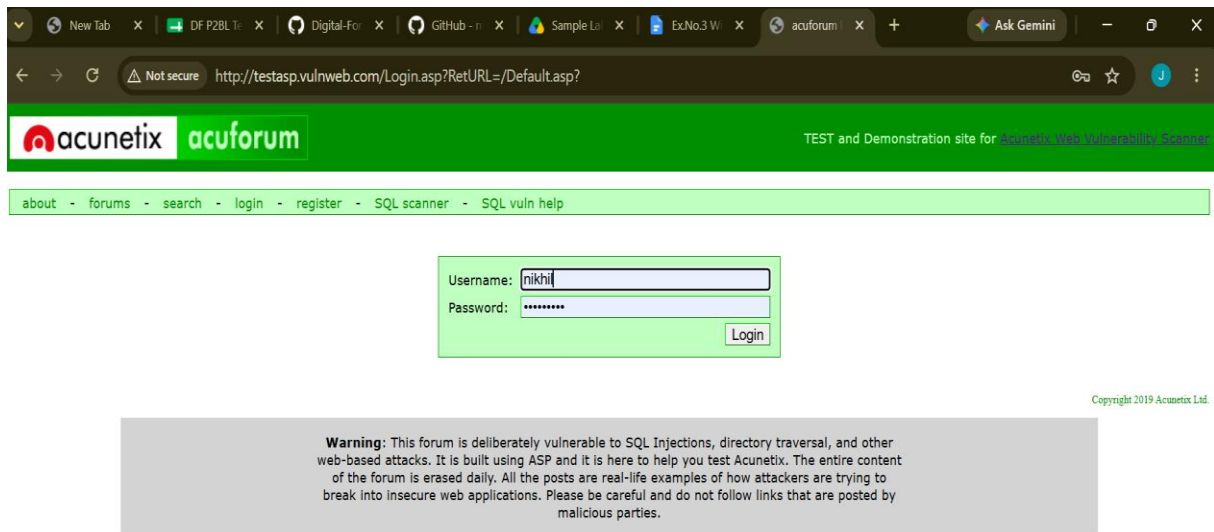
In sniffing can include passwords for various protocols such as HTTP, FTP, Telnet, etc. the captured data can be used to troubleshoot network problems, but can also be used maliciously to gain unauthorized access to sensitive information.

So, here we will see how we can capture the password using the Wireshark network capture analyzer. and see the outputs of the following steps.

Step 1: First of all, open your Wireshark tool in your window or in Linux virtual machine. and start capturing the network. suppose I am capturing my wireless fidelity.



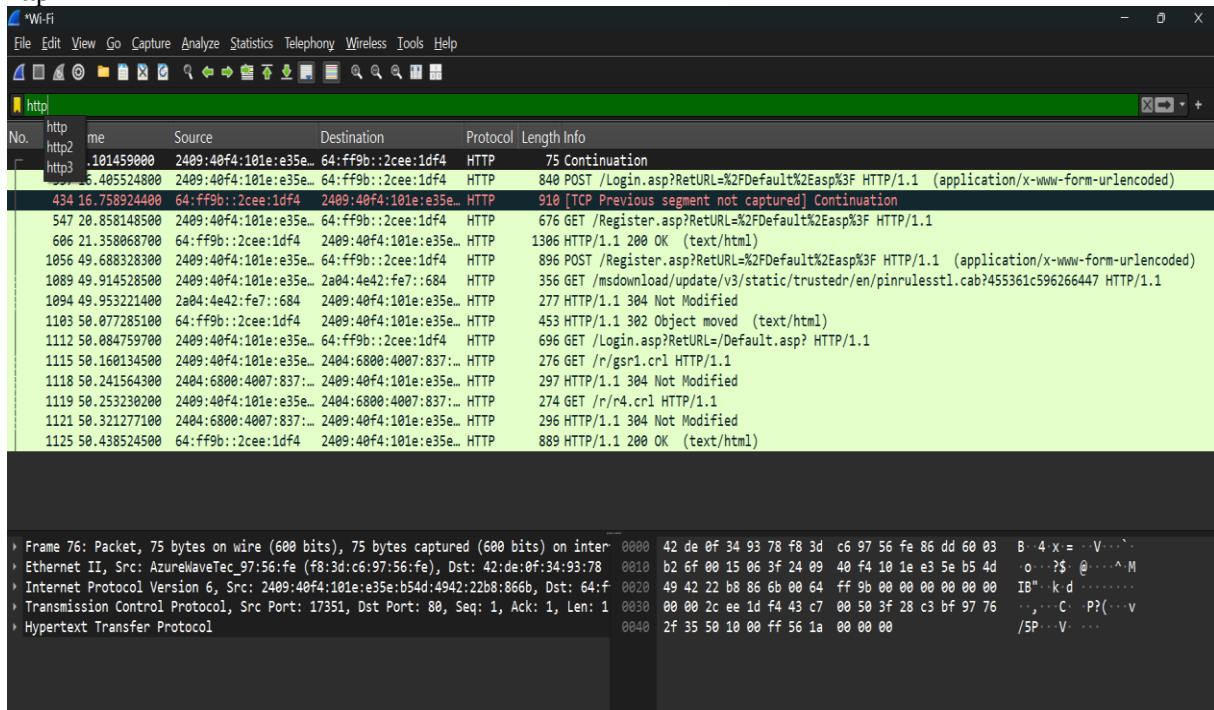
Step 2: After starting the packet capturing we will go to the website and login the credential on that website as you can see in the image.



Step 3: Now after completing the login credential we will go and capture the password in Wireshark. for that we have to use some filter that helps to find the login credential through the packet capturing.

Step 4: Wireshark has captured some packets but we specifically looking for HTTP packets. so in the display filter bar we use some command to find all the captured HTTP packets. as you can see in the below image the green bar where we apply the filter.

http



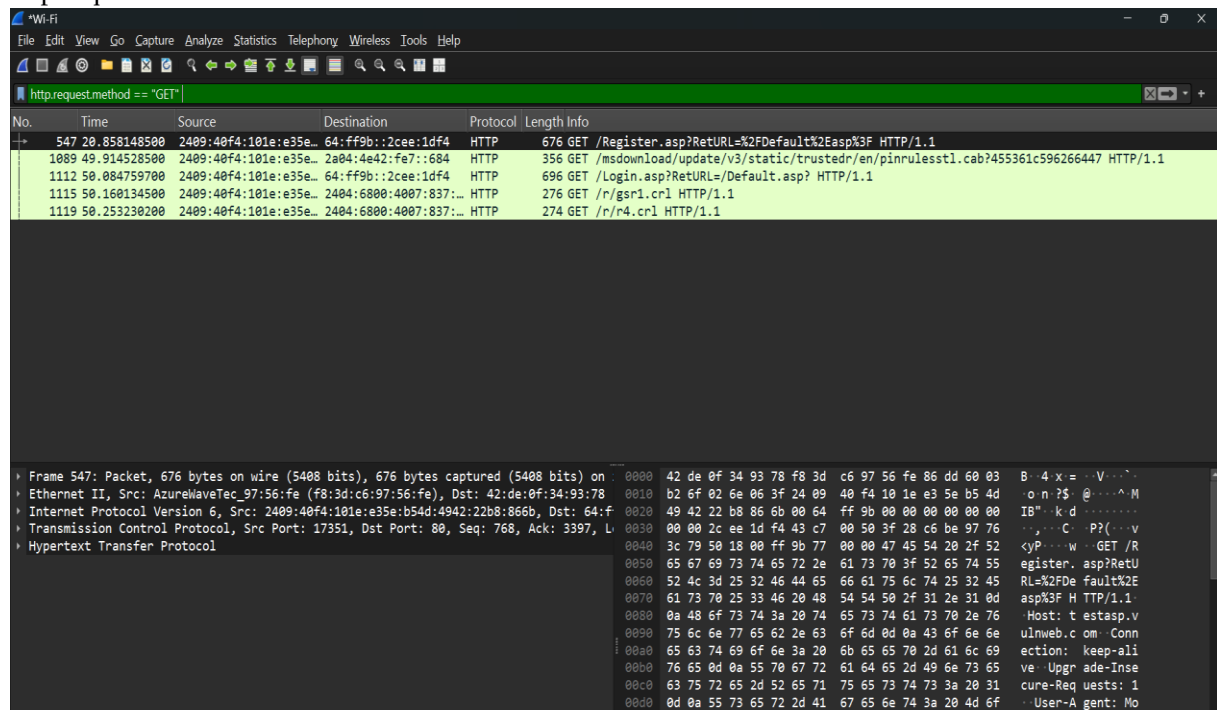
Step 5: So there are some HTTP packets are captured but we specifically looking for form data that the user submitted to the website. for that, we have a separate filter

As we know that there are main two methods used for submitting form data from web pages like login forms to the server. the methods are-

- GET
- POST

Step 6: So firstly for knowing the credential we use the first method and apply the filter for the GET methods as you can see below.

`http.request.method == "GET"`

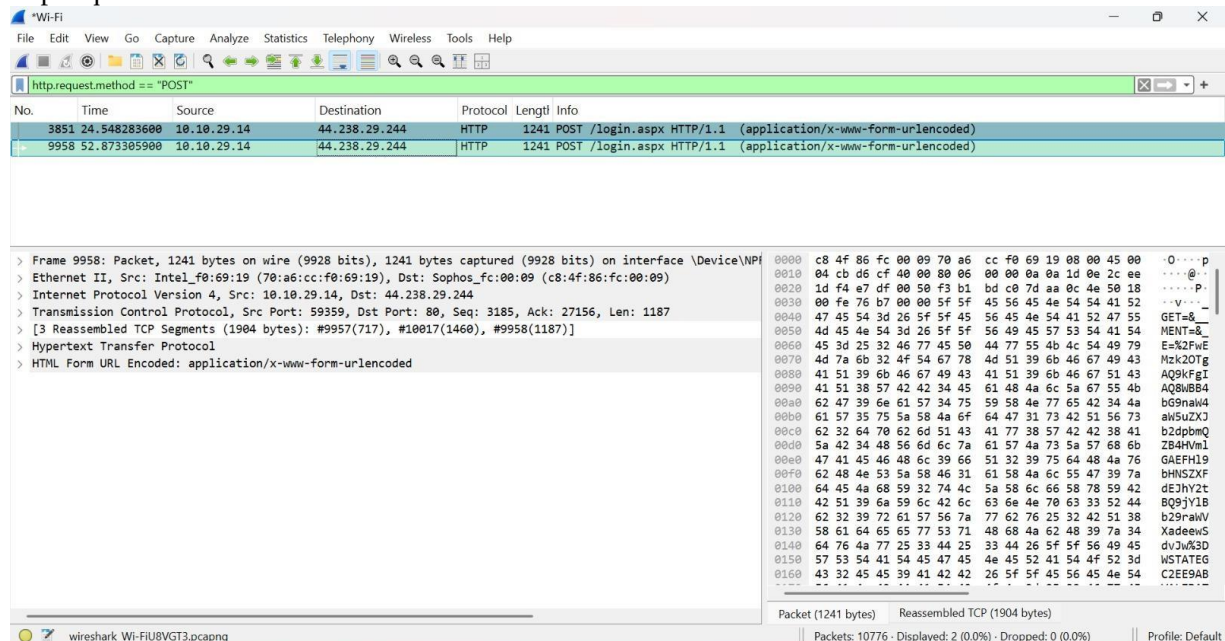


GET method

As you can see in the image there are two packets where the login page was requested with a GET request as well, but there is no form data submitted with a GET request.

Step 7: Now after checking the GET method if we didn't find the form data, then we will try the POST method for that we will apply the filter on Wireshark as you can see.

`http.request.method == "POST"`



As you can see we have a packet with form data click on the packet with user info and the application URL encoded. and click on the down-

HTML form URL Encoded where the login credential is found. login credential as it is the same that we filed on the website in step 2.

Form item: "uname" = "**manideep**"

Form item: "pass" = "**Mani8985**"

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Wireshark - Packet 817 - Wi-Fi

http.request.method

No. Time

817 47.419

Accept-Encoding: gzip, deflate\r\n\r\n

Accept-Language: en-US,en;q=0.9\r\n\r\n

Cookie: ASPSESSIONIDAAAQTSB=EAEIIFJDBLAIIELPKBPIHNL\r\n\r\n

[Response in frame: 818]

Full request URI: http://testasp.vulnweb.com/Login.asp?RetURL=%2Fdefault%2Easp%3F

File Data: 34 bytes

HTML Form URL Encoded: application/x-www-form-urlencoded

Form item: "tfUName" = "nikhil"

Key: tfUName

Value: nikhil

Form item: "tfUPass" = "Nikki@222"

Key: tfUPass

Value: Nikki@222

0260 0a 52 65 66 65 72 65 72 3a 20 68 74 74 70 3a 2f Referer : http:/

0270 2f 74 65 73 74 61 73 70 2e 76 75 6c 6e 77 65 62 /testasp.vulnweb

0280 2e 63 6f 6d 2f 4c 6f 67 69 6e 2e 61 73 70 3f 52 .com/Login.asp?R

0290 65 74 55 52 4c 3d 25 32 46 44 65 66 61 75 6c 74 etURL=%2 Fdefault

02a0 25 32 45 61 73 70 25 33 46 0d 0a 41 63 63 65 70 %2Easp%3 F Accep

02b0 74 2d 45 6e 63 6f 64 69 6e 67 3a 20 67 7a 69 70 t-Encoding: gzip

02c0 2c 20 64 65 66 6c 61 74 65 0d 0a 41 63 63 65 70 , deflate Accep

02d0 74 2d 4c 61 6e 67 75 61 67 65 3a 20 65 6e 2d 55 t-Langua ge: en-U

02e0 53 2c 65 6e 3b 71 3d 30 2e 39 0d 0a 43 6f 6f 6b S,en;q=0 .9 Cook

02f0 69 65 3a 20 41 53 50 53 45 53 53 49 4f 4e 49 44 ie: ASPS ESSIONID

0300 41 41 41 51 54 42 53 42 3d 45 41 45 4c 49 46 4a AAAQTSB =EAEIIFJ

0310 44 42 4c 41 49 49 45 4c 50 4b 42 50 49 48 4e 4c DBLAIIE LPKBPIHNL

0320 45 0d 0a 0d 0a 74 66 55 4e 61 6d 65 3d 6e 69 6b E : : tfU Name=nik

0330 68 69 6c 26 74 66 55 50 61 73 73 3d 4e 69 6b 6b hil&tfUP ass=Nikk

0340 69 25 34 30 32 32 32 1%40222

Value (urlencoded-form.value): 11 bytes

Show packet bytes Layout: Vertical (Stacked)

Close Help

0100 6e 74 65 6e 74 2d 54 79 70 65 3a 20 61 70 70 6c Content-Type: appl

0110 50 63 61 71 69 66 66 3f 70 3d 73 73 3d 66 66 Session / vulnweb